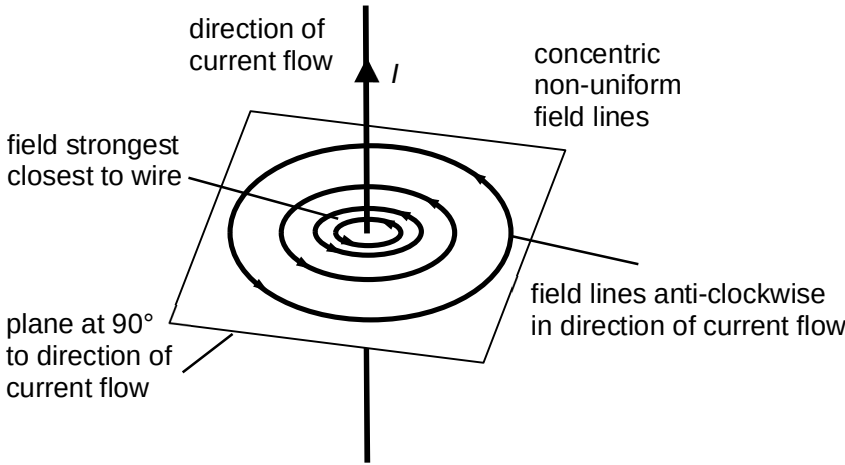
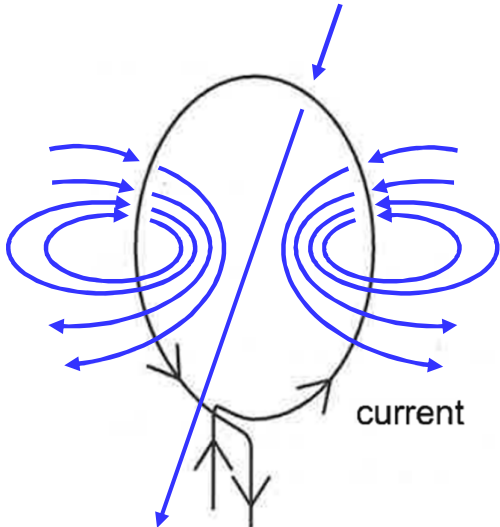
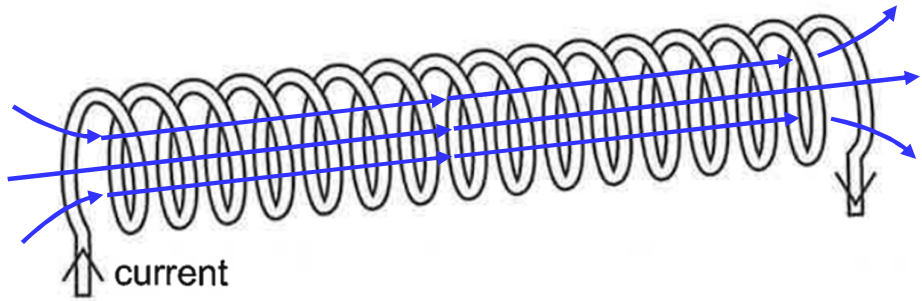


Qn	Solution
1(a)(i)	 direction of current flow I concentric non-uniform field lines field strongest closest to wire plane at 90° to direction of current flow field lines anti-clockwise in direction of current flow
1(a)(ii)	 current
1(a)(iii)	 current

1(b)(i)	The index finger represents the magnetic flux density, which is upward. The second finger represents the current in the copper rod flowing from X to Y. Using Fleming's left-hand rule, the thumb represents the magnetic force acting on the copper rod, which is rightward. Therefore the acceleration of the rod is rightward.
1(b)(ii)	$a = \frac{F}{m} = \frac{(0.56)(2.4)(7.0 \times 10^{-2}) \sin 90^\circ}{0.0052}$ $a = 18.09 \text{ m s}^{-2}$ $a \approx 18 \text{ m s}^{-2}$
2(a)(i)	Geostationary orbit